Jiaan Guo

New York City | jackguo2005@gmail.com | github.com/Jack311045

EDUCATION

New York University, New York City, NY

September 2023 - Present

- **Bachelor of Science** | Joint Major in **Mathematics** and **Data Science**; Minor in **Computer Science** | Expected May 2027
- GPA: 3.6/4.0 | Selected Coursework: Numerical Analysis, Linear & Nonlinear Optimization, Honors Linear Algebra, Machine Learning, Causal Inference, Data Management, Mathematical Statistics, Probability Theory, Algorithms

EXPERIENCE

SUSTENTRA - *Data Science & Engineering Intern*

June 2026 - September 2026 | Remote

- Built a Python document-intelligence pipeline using AWS S3, Amazon Textract, and the OpenAI and Gemini API to process 40+ client-provided ESG documents averaging more than 10 pages and hundreds of extracted fields each; achieved over 99% client-validated field-level extraction accuracy, with source snippets for nearly all fields.
- Parsed documents in 30 seconds or less before secondary API validation and achieved a 98% first-pass JSON schema-validation rate, with custom validation code and API-based checks enforcing final schema conformance.
- Translated NY Part 253, ISSA 5000, and ISO 14064 requirements into versioned JSON schemas; integrated 90 formula records and 63 combustion emission factors to support calculation and reconciliation workflows.
- Developed a seven-stage Streamlit audit prototype and authored 170+ passing schema, UI, and regression tests that identified and resolved five major regression and schema issues; presented the system to 10+ audit-industry and client representatives, with selected features incorporated into a React application.

GIGACLOUD Technology - *Data Science Intern*

July 2025 - August 2025 | Suzhou, China

- Analyzed 100,000+ shipment records and found surcharge exposure in approximately one-third of shipments, with girth- and longest-side-related charges emerging as the two leading drivers, particularly for parcels near carrier thresholds.
- Engineered geometry and carrier-rule features to segment subsequently surcharged shipments into pre-shipment measurement bands aligned with carrier surcharge thresholds; girth- and longest-side-related cases represented approximately 80% to 90% of the grouped cases, with a roughly even split between the two categories.

SELECTED PROJECTS

ChessLens | Python, PyTorch, ONNX, DuckDB, dbt, SQL, Docker, AWS (ECS/CloudWatch), MLflow, GitHub Actions, FastAPI

May 2026 - Current

- Built a streaming Lichess PGN data pipeline with deterministic IDs, HMAC anonymization, sharded Parquet outputs, resumable manifests, and staged publishing; completed and reconciled 43 ingestion shards covering 718M+ moves.
- Validated a deterministic 10% analysis sample across 8 dbt models and 39 data tests (47/47 nodes successful); added typed CLI and idempotency checks with passing Ruff, mypy, unit, property, and integration tests.

Recipe Discovery & Semantic Search | Python, PyTorch, Sentence Transformers, SigLIP, TF-IDF, Streamlit

Jan 2026 - May 2026

- Built an end-to-end recipe discovery application on the Food.com Recipes and Interactions dataset, combining SBERT dense embeddings with TF-IDF for hybrid text retrieval and SigLIP for image and combined image-text search; supported natural-language queries and dietary filtering.
- Implemented k-means from scratch in PyTorch for 10,000+ recipe embeddings, using k-means++ initialization, Lloyd updates, empty-cluster reseeding, and multiple restarts to produce 30 clusters with a 0.72 silhouette score on the full dataset; visualized cluster structure with PCA and autoencoder projections.

Met Museum Cross-Modal Search | Python, PyTorch, FAISS, FastAPI, Streamlit

Dec 2025 - Jan 2026

- Built a cross-modal retrieval system for the Metropolitan Museum of Art Open Access dataset, supporting text-to-image and image-to-text search through a shared embedding space. Combined DINOv2 image embeddings, Sentence Transformers text embeddings, dual-encoder training, and FAISS indexing into an end-to-end retrieval pipeline with CLI, FastAPI, and Streamlit interfaces.

SKILLS

- Languages: Python, SQL, Java, C, C++
- Machine Learning: PyTorch, ONNX, scikit-learn, SciPy, statsmodels, FAISS, LLM APIs (OpenAI, Gemini)
- Data Engineering & BI: Excel, pandas, NumPy, DuckDB, dbt, Parquet, JSON Schema, PowerBI, ETL/Testing
- Tools & MLOps: AWS (S3, Textract, ECS, CloudWatch), Docker, FastAPI, Streamlit, Linux, Git, pytest, CI/CD